

Introduction to Probability and Statistics I

Lecture 2: Data organisation

PETER NYANDIRI

Strathmore Institute of Mathematical Sciences

Overview

- 1 Learning Outcomes
- 2 Arranging Quantitative Data
- 3 Terminologies
- 4 Graphical Representation of quantitative data
- 5 Arranging Qualitative Data
- 6 Graphical Representation of qualitative data

Learning Outcomes

- Arrange qualitative data
 - i. Simple tables
 - ii. Cross tabulation
- Arrange quantitative data
 - i. Simple arrays
 - ii. Frequency tables
- Present qualitative data
 - i. Bar charts
 - ii. Pie charts
- Present quantitative data
 - i. Histogram
 - ii. Line graphs

Simple arrays

Is an arrangement of data in an ascending or descending order.
Its convenient if the number of its small.

Example

The following data represents monthly wages of employees (in US dollars) in Company B.

43, 82, 42, 20, 53, 22, 50, 22, 28, 22, 20, 12, 25, 14

A simple array will look like this

12, 14, 20, 20, 22, 22, 22, 25, 28, 42, 43, 50, 53, 82

Frequency Distribution

◇ **Frequency distribution**

A frequency distribution is a table that lists categories of items along with their corresponding frequencies.

◇ **Guidelines for constructing a frequency distribution table**

- * The classes must be mutually exclusive. That is, each score must belong to exactly one class.
- * The classes must be exhaustive meaning there should be enough classes to accommodate all data
- * All classes should have the same width
- * The number of classes should be between 5 and 20.

An example

- **Example:** Consider the guessed weights collected in our first class from 40 students.

140 135 140 160 175 150 152 155 155 165
160 170 155 140 160 160 175 140 145 150
145 155 150 150 165 148 152 155 155 160
165 170 160 140 150 150 152 155 130 155

- **Problem:** Let us organize it into a frequency distribution table.

Procedure for Constructing a frequency distribution

① Sort the data - Create a frequency array

② Determine the Range (R)

$$\text{Range (R)} = \text{Largest values} - \text{Smallest value} = 175 - 130 = 45$$

③ Choose the number of classes (K):

$$k = 1 + 3.322 * \log(N) \text{ or } 1 + \frac{\log(N)}{\log(2)}$$

Round up the answer to the next integer.

$$\text{E.g. } k = 1 + 3.322 * \log(40) = 6.322 \simeq 6$$

④ Determine the class width (W)

$$W = \frac{R}{K} = 45/6 = 7.11 \simeq 7$$

- 1 Write down your lowest value; The lowest value is 130
- 2 Add the class width from Step 4 to Step 5 to get the next lower class limit: $130 + 7 = 137$
- 3 Repeat Step 6 for the other minimum data values until you have created the number of classes you chose in Step 1. We chose 7 classes, so our 7 minimum data values are: 118

137 ($130+7$)

144 ($137+7$)

151 ($144+7$)

158 ($151+7$)

165 ($158+7$)

172 ($165+7$)

179 ($172+7$)

- 1 Write down the upper class limits. For example:

$$130 + (7-1) = 136 \quad \longrightarrow \quad 130 - 136$$

$$137 + (7-1) = 143 \quad \longrightarrow \quad 137 - 143$$

$$144 + (7-1) = 150 \quad \longrightarrow \quad 144 - 150$$

$$151 + (7-1) = 157 \quad \longrightarrow \quad 151 - 157$$

$$158 + (7-1) = 164 \quad \longrightarrow \quad 158 - 164$$

$$165 + (7-1) = 171 \quad \longrightarrow \quad 165 - 171$$

$$172 + (7-1) = 178 \quad \longrightarrow \quad 172 - 178$$

- 1 Add a second column for the number of items in each class, and label the columns with appropriate headings:

Weight	Frequency
130 - 136	2
137 - 143	5
144 - 150	9
151 - 157	11
158 - 164	6
165 - 171	5
172 - 178	2

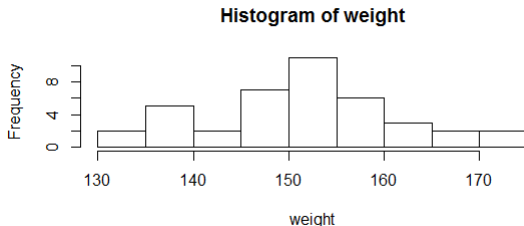
Terminologies associated with the table

- ◇ **Grouped data:** Data organized into a frequency distribution table
- ◇ **Class frequency:** The number of observations in each class.
- ◇ **Class relative frequency:** The percent of observations in each class.
- ◇ **Class cumulative frequency:** The total observations up to certain class
- ◇ **Class Midpoint:** A point that divides a class into two equal parts, i.e. the average of the upper and lower class limits values.
- ◇ **Class interval:** The difference between two consecutive lower class limits

Weight	F	RF	CF	mid point	Class boundaries
130 - 136	2	0.05	2	133	129.5 - 136.5
137 - 143	5	0.125	7	140	136.5 - 143.5
144 - 150	9	0.225	16	147	143.5 - 150.5
151 - 157	11	0.275	27	154	150.5 - 157.5
158 - 164	6	0.15	33	161	157.5 - 164.5
165 - 171	5	0.125	38	168	164.5 - 171.5
172 - 178	2	0.05	40	175	171.5 - 178.5

Histogram

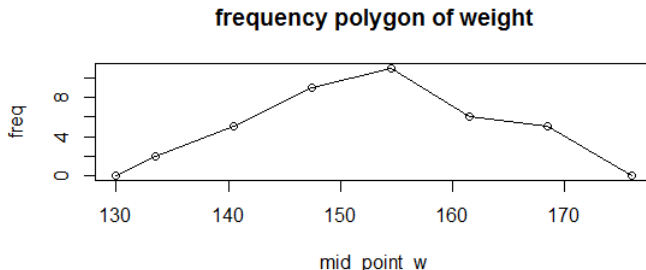
- **Histogram:** Displays the data by using continuous vertical bars of various heights to represent the frequencies of the classes.
- histogram for the weight example



- R code for the histogram; `> hist(weight)`

Frequency polygon

- **Frequency polygon:** Consists of line segments connecting the points formed by the class midpoint and the class frequency.
- Frequency polygon of weight



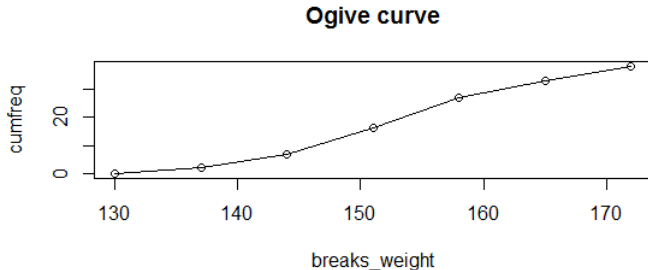
Frequency polygon

Frequency polygon - R code:

```
> m <- -min(weight)
> M <- -max(weight) + 1
> breaks_weight <- -seq(m, M, by = 7)
> weight.cut <- -cut(weight, breaks_weight, right = FALSE)
> weight.freq <- -table(weight.cut)
> freq <- -c(0, weight.freq, 0)
> midpoint <- -seq(m + 3.5, M - 3.5, by = 7)
> midpoint <- -c(m, midpoint, M)
> plot(midpoint, freq, main = "frequency polygon of weight")
> lines(midpointw, freq)
```

Ogive: cumulative frequency polygon

- **Cumulative frequency polygon:** The ogive is a graph that represents the cumulative frequencies for the classes in a frequency distribution..
- Ogive for the weight example



cumulative frequency polygon

Ogive curve - R code:

```
> lines(breaks_w eight, cumfreq)
> m <- -min(weight)
> M <- -max(weight) + 1
> breaks_w eight <- -seq(m, M, by = 7)
> weight.cut <- -cut(weight, breaks_w eight, right = FALSE)
> weight.freq <- -table(weight.cut)
> weight.cumfreq <- -cumsum(weight.freq)
> cumfreq = c(0, weight.cumfreq)
> plot(breaks_w eight, cumfreq, main = " Ogivecurve" )
> lines(breaks_w eight, cumfreq)
```

Tabulations

- A table is a systematic arrangement of statistical data in columns and rows
- Rows are horizontal arrangements while columns are vertical arrangements
- The purpose of a table is to simplify the presentation and to facilitate comparison

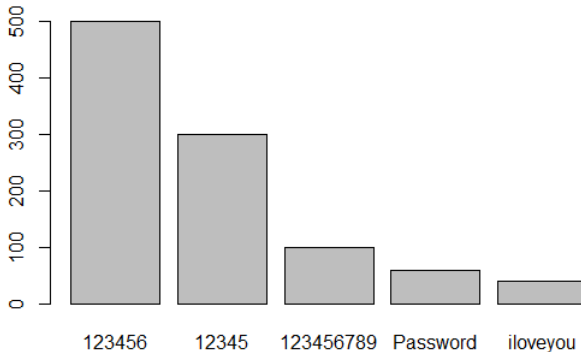
Simple tables

- **Simple table:** One way table
- **Example:** Here are the five worst passwords: a study shows the following result among 1,000 persons surveyed:

Password	number of Users
123456	500
12345	300
123456789	100
Password	60
iloveyou	40

Bar chart

Types of password Bar chart

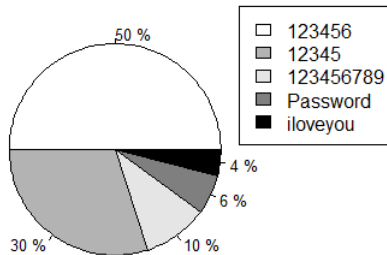


R code for Bar chart

```
> password =  
c('123456',' 12345',' 123456789',' Password',' iloveyou')  
> numberofusers = c(500, 300, 100, 60, 40)  
> pw <- data.frame(password, numberofusers)  
> attach(pw)  
> barplot(numberofusers, password, main =  
" TypesofpasswordBarchart" )
```

Pie chart

Types of password



R code for Pie chart

```
> #piechart
> #Calculate the percentage for each category.
> numberofusers_label <- round(numberofusers/sum(numberofusers)*100,
> #This adds a space and a percent sign.
> numberofusers_label <- paste(numberofusers_label, "%", sep=" ")
> colors <- c("white","grey70","grey90","grey50","black")
> pie(numberofusers, main="Types of password",labels=numberofusers_label,
0.8)
> legend("topright", c('123456', '12345', '123456789', 'Password', 'ilove'),
```

Exercise

Grams per Food Serving

The data shown are the number of grams per serving of 30 selected brands of cakes. Construct a frequency distribution

32 47 51 41 46 30

46 38 34 34 52 48

48 38 43 41 21 24

25 29 33 45 51 32

32 27 23 23 34 35

If grams above 30 was considered excess, what percentage was not excess